

ZHONGHAO LIN

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EDUCATION

Beijing Institute of Technology (BIT), Beijing, China

Sept. 2024 – Jun. 2027 (Expected)

M.ENG. in CONTROL ENGINEERING

GPA: 92.40 / 100

Selected Graduate Coursework: Linear Algebra in Automatic Control (97 / 100), Optimization Theory and Methods (96 / 100), Multi-Source Information Filtering and Fusion (95 / 100), Multi-Agent Coordination and Control (94 / 100), Linear System Theory (92 / 100)

Advisor: Dr. Xianlin Zeng

Research Topics: Time-varying optimization, online optimization, distributed optimization, and their applications to localization, tracking, and networked systems.

Beijing Institute of Technology (BIT), Beijing, China

Sept. 2020 - Jun. 2024

B.ENG. in AUTOMATION

GPA: 91.52 / 100 (ranked top 3%)

Selected Coursework: Mathematical Analysis (99 / 100), Linear Algebra (99 / 100), Complex Functions (99 / 100), Operations Research (99 / 100), Probability Theory (98 / 100), Intelligent Control (97 / 100), Automatic Control Theory (96 / 100), Information Theory (95 / 100), Signal Analysis and Processing (94 / 100).

PUBLICATIONS AND MANUSCRIPTS

1. **Z. Lin**, J. Hou, and X. Zeng, “Optimal Prediction-Correction Algorithm Using Sparse Linear Extrapolation for Time-Varying Optimization,” *IEEE Transactions on Signal Processing*, vol. 74, pp. 2564-2578, 2026.
2. **Z. Lin**, X. Zeng, J. Hou, J. Sun, and J. Chen, “Primal-Dual Prediction-Correction Method with Tunable Memory for Linearly Constrained Time-Varying Convex Optimization,” *Journal of Systems Science and Complexity*, vol. 39, no. 2, pp. 483–510, 2026.
3. **Z. Lin**, J. Hou, and X. Zeng, “Hessian-Inverse-Free Prediction-Correction Method for Time-Varying Convex Optimization,” accepted in *IEEE Conference on Decision and Control (CDC 2026)*.
4. S. Wang, **Z. Lin**, X. Zeng, “Distributed Optimization with Anytime Coupling Constraints and Asymptotic Consensus,” accepted in *International Conference on Automation, Control and Robotics (ICACR 2026)*.
5. **Z. Lin**, “Linear-Extrapolation Prediction-Correction for TOA-Based Target Tracking,” working paper.

RESEARCH EXPERIENCE

Prediction-Correction Methods for Time-Varying Optimization (Master’s thesis research)

Advisor: Dr. Xianlin Zeng, BIT, China

Sept. 2024 – Present

- Proposed a sparse linear extrapolation prediction-correction (SE-PC) method for strongly convex time-varying optimization, where predictor coefficients are designed via ℓ_1 -norm minimization under tractable constraints. The method attains state-of-the-art tracking accuracy while substantially reducing the number of gradient-descent correction steps required for convergence. This work has been published in *IEEE Transactions on Signal Processing*.
- Developed a primal-dual prediction-correction (PD-PC) method with tunable memory for linearly constrained time-varying convex optimization. The method combines tunable-memory extrapolation with primal-dual gradient correction and achieves second-order asymptotic tracking accuracy under time-varying linear constraints. This work has been published in *Journal of Systems Science and Complexity*.
- Proposed a Hessian-inverse-free prediction-correction flow for continuous-time time-varying convex optimization. The method introduces an auxiliary matrix dynamics to generate an online surrogate of the Hessian inverse, thereby avoiding explicit matrix inversion. This work is currently accepted for presentation at the *IEEE Conference on Decision and Control (CDC 2026)*.
- Extended the linear-extrapolation prediction-correction approach from the previous TSP work to TOA-based target tracking. Established that both extrapolated predictions and corrected iterates remain within

the local strong-convexity region of the current TOA least-squares objective. Derived an explicit tracking bound showing Q -linear convergence to a noise-dependent neighborhood, which vanishes in the noiseless case. This manuscript is being prepared for future submission to *IEEE Signal Processing Letters*.

Distributed Optimization-Based Cooperative Velocity Control (Bachelor’s thesis research)

Advisor: Dr. Xianlin Zeng, BIT, China Sept. 2023 – Jun. 2024

- Formulated the cooperative velocity control problem of multiple autonomous vehicles as a consensus-based distributed optimization problem, where fuel consumption was modeled as the objective function and safety and communication requirements were incorporated as time-varying constraints.
- Developed a distributed velocity control algorithm by combining projected gradient methods, control barrier functions, and sliding-mode control, and validated its convergence and robustness through multi-vehicle simulations.

SELECTED HONORS AND AWARDS

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| 1. Special-Class Graduate Scholarship (ranked top 5%), BIT. | 2024 |
| 2. Outstanding Graduate , BIT. | 2024 |
| 3. Samsung Scholarship (ranked 3rd among 200+ students), China Samsung. | 2023 |
| 4. First Prize , Chinese Mathematics Competitions for College Students, awarded twice. | 2021, 2022 |
| 5. First-Class Undergraduate Scholarship (ranked top 5%), BIT, awarded four times. | 2021–2024 |
| 6. First Prize , BIT Freshman Mathematics Competition. | 2020 |

ACADEMIC SERVICE

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| 1. Reviewer for <i>IEEE International Conference on Control and Automation (ICCA 2026)</i> . | 2026 |
| 2. Reviewer for <i>Chinese Control Conference (CCC 2026)</i> . | 2026 |

SKILLS

- **Languages:** Mandarin Chinese (native), English (CET-6: 580 / 710, IELTS: 7.0).
- **Programming and Tools:** MATLAB, python, $\LaTeX$, Vibe coding.

APPENDIX: PUBLICATION AND MANUSCRIPT ABSTRACTS

1. **Representative Work:** Sparse Linear Extrapolation Prediction-Correction for Time-Varying Optimization

Publication: Z. Lin, J. Hou, and X. Zeng, “Optimal Prediction-Correction Algorithm Using Sparse Linear Extrapolation for Time-Varying Optimization,” *IEEE Transactions on Signal Processing*, vol. 74, pp. 2564–2578, 2026.

Abstract. This paper introduces an optimal prediction-correction algorithm leveraging sparse linear extrapolation for strongly convex, unconstrained time-varying optimization problems, which are prevalent in dynamic systems and online learning. The proposed method constructs the prediction phase as a sparse linear combination of past iterates, with extrapolation coefficients derived by solving an ℓ_1 -norm minimization problem under tractable constraints. By promoting sparsity in the predictor, the algorithm reduces the frequency of correction steps and the associated computational cost of gradient evaluations. We establish the existence of an ℓ_1 -optimal sparse predictor and derive closed-form solutions for second- and third-order tracking accuracy cases. Theoretical analysis confirms that the method achieves state-of-the-art tracking accuracy with improved computational efficiency compared to existing prediction-correction approaches. Numerical experiments validate the theoretical results, demonstrating the advantages of the proposed algorithm in reducing computational overhead while maintaining high accuracy.

2. Primal-Dual Prediction-Correction for Linearly Constrained Time-Varying Optimization

Publication: Z. Lin, X. Zeng, J. Hou, J. Sun, and J. Chen, “Primal-Dual Prediction-Correction Method with Tunable Memory for Linearly Constrained Time-Varying Convex Optimization,” *Journal of Systems Science and Complexity*, vol. 39, no. 2, pp. 483–510, 2026.

Abstract. This paper presents a primal-dual prediction-correction (PD-PC) method for solving linearly constrained time-varying convex optimization problems, which frequently arise in control, signal processing, and online learning applications. The proposed method establishes a novel integration of primal-dual gradient dynamics with a discrete-time prediction-correction structure, specifically designed for problems with time-dependent linear constraints. A tunable memory parameter is introduced in the prediction phase to perform linear extrapolation using past iterates, enabling a flexible trade-off between the amount of historical information stored and the computational cost of correction. In the correction phase, primal and dual variables are updated via gradient descent-ascent iterations, thus maintaining the computational efficiency of a first-order method without requiring Hessian or high-order derivative computations. Theoretical analysis shows that the method achieves $\mathcal{O}(h^2)$ asymptotic tracking accuracy for both primal and dual variables, matching the state-of-the-art performance among first-order methods even in unconstrained settings. Numerical experiments on problems with both time-invariant and time-varying constraints validate the theoretical findings and demonstrate the method’s effectiveness.

3. Prediction-Correction Dynamics for Continuous-Time Time-Varying Convex Optimization

Manuscript: Z. Lin, J. Hou, and X. Zeng, “Hessian-Inverse-Free Prediction-Correction Method for Time-Varying Convex Optimization,” accepted for presentation at the *IEEE Conference on Decision and Control (CDC 2026)*.

Abstract. In this paper, we propose a Hessian-inverse-free prediction-correction method for time-varying convex optimization. The proposed method embeds an auxiliary H flow into a prediction-correction x flow, where the H flow generates an online surrogate of the Hessian inverse through prediction, correction, and anchoring terms, thereby avoiding explicit matrix inversion. Under uniform strong convexity, the coupled x – H flow guarantees convergence of the gradient norm to zero and asymptotic tracking of the time-varying optimal trajectory. In the merely convex case, the proposed method remains well-posed and yields a tunable asymptotic bound on the gradient norm. A numerical example illustrates the efficacy of the proposed method.

4. Linear-Extrapolation Prediction-Correction for TOA-Based Target Tracking

Manuscript (draft): Z. Lin, “Linear-Extrapolation Prediction-Correction for TOA-Based Target Tracking,” working paper; planned for submission to *IEEE Signal Processing Letters*.

Abstract (draft). This letter studies target tracking from sequential time-of-arrival (TOA) measurements. The problem is formulated as a time-varying TOA least-squares problem, whose objective is generally nonconvex and locally strongly convex only near its least-squares estimate. Motivated by polynomial and spline trajectory models used in autonomous systems, we propose a linear-extrapolation prediction-correction approach for TOA-based tracking. The method predicts the next target position from past iterates and then applies a few gradient correction steps to the current TOA loss. We prove that, under standard local TOA regularity conditions, both the extrapolated predictions and the corrected iterates remain inside the local strong-convexity region of the current TOA least-squares objective, leading to an explicit tracking bound with respect to the true target trajectory. The bound further shows that the tracking error converges Q -linearly to a noise-dependent neighborhood, which reduces to zero in the noiseless case.